In [1]:
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# ==========================================================
# 3. AUDIO CLASSIFICATION NOTEBOOK
# ==========================================================

# ------------------------------------------------
# 3.1 IMPORTS AND SETUP
# ------------------------------------------------
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import os
import librosa
import librosa.display
from sklearn.model_selection import train_test_split, cross_val_score, StratifiedKF
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.ensemble import RandomForestClassifier
from sklearn.svm import SVC
from sklearn.metrics import classification_report, confusion_matrix, accuracy_score
from sklearn.decomposition import PCA
import warnings
warnings.filterwarnings('ignore')

print("Libraries imported successfully")

# ------------------------------------------------
# 3.2 LOAD DATA
# ------------------------------------------------
print("\nLoading dataset...")
# Use relative path to locate the dataset file
data_path = "data/overview-of-recordings.csv"
data = pd.read_csv(data_path)
print(f"Dataset shape: {data.shape}")
print(f"Number of unique prompts: {data['prompt'].nunique()}")
print(f"Prompt classes: {data['prompt'].unique()}")

# ------------------------------------------------
# 3.3 AUDIO FEATURE EXTRACTION
# ------------------------------------------------
# Define function to extract audio features
def extract_features(file_path):
    try:
        # Load audio file
        y, sr = librosa.load(file_path, sr=None)

        # Extract features
        # Time-domain features
        duration = librosa.get_duration(y=y, sr=sr)
        zero_crossing_rate = np.mean(librosa.feature.zero_crossing_rate(y))
        zero_crossing_rate_std = np.std(librosa.feature.zero_crossing_rate(y))
        rms = np.mean(librosa.feature.rms(y=y))
        rms_std = np.std(librosa.feature.rms(y=y))
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        # Frequency-domain features
        spectral_centroid = np.mean(librosa.feature.spectral_centroid(y=y, sr=sr))
        spectral_centroid_std = np.std(librosa.feature.spectral_centroid(y=y, sr=sr
        spectral_bandwidth = np.mean(librosa.feature.spectral_bandwidth(y=y, sr=sr)
        spectral_bandwidth_std = np.std(librosa.feature.spectral_bandwidth(y=y, sr=
        spectral_rolloff = np.mean(librosa.feature.spectral_rolloff(y=y, sr=sr))
        spectral_rolloff_std = np.std(librosa.feature.spectral_rolloff(y=y, sr=sr))

        # Mel-frequency cepstral coefficients
        mfccs = librosa.feature.mfcc(y=y, sr=sr, n_mfcc=13)
        mfccs_mean = np.mean(mfccs, axis=1)
        mfccs_std = np.std(mfccs, axis=1)

        # Tempo
        tempo, _ = librosa.beat.beat_track(y=y, sr=sr)

        # Chroma features
        chroma = librosa.feature.chroma_stft(y=y, sr=sr)
        chroma_mean = np.mean(chroma, axis=1)
        chroma_std = np.std(chroma, axis=1)

        # Combine all features into a flat array
        feature_list = [
            duration, zero_crossing_rate, zero_crossing_rate_std,
            rms, rms_std, spectral_centroid, spectral_centroid_std,
            spectral_bandwidth, spectral_bandwidth_std,
            spectral_rolloff, spectral_rolloff_std,
            tempo
        ]

        # Add MFCCs
        feature_list.extend(mfccs_mean)
        feature_list.extend(mfccs_std)

        # Add chroma features
        feature_list.extend(chroma_mean)
        feature_list.extend(chroma_std)

        return feature_list

    except Exception as e:
        print(f"Error extracting features from {file_path}: {e}")
        return None

# Define feature names for later use
feature_names = [
    'duration', 'zero_crossing_rate', 'zero_crossing_rate_std',
    'rms', 'rms_std', 'spectral_centroid', 'spectral_centroid_std',
    'spectral_bandwidth', 'spectral_bandwidth_std',
    'spectral_rolloff', 'spectral_rolloff_std',
    'tempo'
]

# Add MFCC feature names
for i in range(1, 14):
    feature_names.append(f'mfcc{i}_mean')
```

```
for i in range(1, 14):
    feature_names.append(f'mfcc{i}_std')

# Add chroma feature names
for i in range(1, 13):
    feature_names.append(f'chroma{i}_mean')
for i in range(1, 13):
    feature_names.append(f'chroma{i}_std')

print(f"\nTotal number of audio features: {len(feature_names)}")

# Define base audio directory
audio_base_dir = "data/recordings/"

# Function to find audio file in the correct subdirectory
def find_audio_file(file_name):
    # Check each subdirectory
    for subdir in ['train', 'test', 'validate']:
        file_path = os.path.join(audio_base_dir, subdir, file_name)
        if os.path.exists(file_path):
            return file_path, subdir
    return None, None

# Check if audio directory exists
if not os.path.exists(audio_base_dir):
    print(f"Warning: Audio directory {audio_base_dir} not found!")
    print("Creating a dummy audio directory structure for demonstration")

    # Create dummy directory structure
    for subdir in ['train', 'test', 'validate']:
        os.makedirs(os.path.join(audio_base_dir, subdir), exist_ok=True)

    print("Please ensure your audio files are correctly placed in the recordings di

# Extract features for each audio file
print("\nExtracting audio features...")
audio_features = []
audio_labels = []
audio_ids = []
audio_splits = []

# Process files in batches to show progress
total_files = len(data)
batch_size = 50  # Process this many files before reporting progress
processed = 0

for i, row in data.iterrows():
    file_path, split = find_audio_file(row['file_name'])

    if file_path:
        features = extract_features(file_path)

        if features:
            audio_features.append(features)
            audio_labels.append(row['prompt'])
            audio_ids.append(row['file_name'])
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            audio_splits.append(split)

    # Print progress
    processed += 1
    if processed % batch_size == 0 or processed == total_files:
        print(f"Processed {processed}/{total_files} files")

# Check if we have any features
if len(audio_features) == 0:
    print("\nNo audio features extracted. Creating dummy data for demonstration.")

    # Create dummy data for demonstration
    feature_count = len(feature_names)
    sample_count = min(100, len(data))

    # Generate random features
    np.random.seed(42)
    audio_features = np.random.randn(sample_count, feature_count)

    # Sample prompts from the dataset
    indices = np.random.choice(len(data), sample_count, replace=False)
    audio_labels = data['prompt'].iloc[indices].values
    audio_ids = data['file_name'].iloc[indices].values

    # Assign dummy splits
    splits = ['train', 'test', 'validate']
    audio_splits = np.random.choice(splits, sample_count)

    print(f"Created {sample_count} dummy audio samples for demonstration.")
else:
    print(f"\nSuccessfully extracted features from {len(audio_features)} audio file

# Convert to numpy arrays
X = np.array(audio_features)
y = np.array(audio_labels)

# Create a DataFrame for easier analysis
audio_df = pd.DataFrame(X, columns=feature_names)
audio_df['prompt'] = y
audio_df['file_name'] = audio_ids
audio_df['split'] = audio_splits

print(f"Audio features dataframe shape: {audio_df.shape}")

# ------------------------------------------------
# 3.4 EXPLORATORY DATA ANALYSIS OF AUDIO FEATURES
# ------------------------------------------------
print("\nPerforming exploratory analysis of audio features...")

# Basic statistics of audio features
print("\nBasic statistics of selected audio features:")
selected_features = ['duration', 'rms', 'spectral_centroid', 'zero_crossing_rate',
print(audio_df[selected_features].describe())

# Distribution of audio features by prompt
plt.figure(figsize=(15, 10))
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for i, feature in enumerate(selected_features, 1):
    plt.subplot(2, 3, i)
    sns.boxplot(x='prompt', y=feature, data=audio_df)
    plt.title(f'{feature} by Prompt')
    plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# Feature correlations
print("\nCalculating feature correlations...")
corr_matrix = audio_df[feature_names].corr()

# Plot correlation heatmap (only for a subset of features to keep it readable)
plt.figure(figsize=(12, 10))
main_features = [
    'duration', 'rms', 'spectral_centroid', 'spectral_bandwidth',
    'spectral_rolloff', 'zero_crossing_rate', 'tempo',
    'mfcc1_mean', 'mfcc2_mean', 'mfcc3_mean', 'mfcc4_mean', 'mfcc5_mean'
]
sns.heatmap(corr_matrix.loc[main_features, main_features], annot=True, cmap='coolwa
plt.title("Correlation Between Main Audio Features")
plt.tight_layout()
plt.show()

# PCA for visualizing high-dimensional audio features
print("\nPerforming PCA for dimensionality reduction...")

# Standardize features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Apply PCA
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

# Create a DataFrame for plotting
pca_df = pd.DataFrame({
    'PC1': X_pca[:, 0],
    'PC2': X_pca[:, 1],
    'prompt': y
})

# Plot PCA results
plt.figure(figsize=(10, 8))
sns.scatterplot(x='PC1', y='PC2', hue='prompt', data=pca_df, palette='Set1')
plt.title('PCA of Audio Features')
plt.xlabel(f'PC1 ({pca.explained_variance_ratio_[0]:.2%} variance)')
plt.ylabel(f'PC2 ({pca.explained_variance_ratio_[1]:.2%} variance)')
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()

# Variance explained by each principal component
plt.figure(figsize=(10, 6))
plt.bar(range(1, min(11, len(pca.explained_variance_ratio_))+1),
        pca.explained_variance_ratio_[:10], alpha=0.5, align='center')
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```
plt.step(range(1, min(11, len(pca.explained_variance_ratio_))+1),
         np.cumsum(pca.explained_variance_ratio_[:10]), where='mid')
plt.ylabel('Explained variance ratio')
plt.xlabel('Principal component index')
plt.title('Explained Variance by Principal Components')
plt.tight_layout()
plt.show()

# ------------------------------------------------
# 3.5 DATA SPLITTING FOR TRAINING AND TESTING
# ------------------------------------------------
print("\nSplitting data into training and testing sets...")

# Check if we already have train/test splits
if 'split' in audio_df.columns and len(audio_df['split'].unique()) > 1:
    print("Using existing train/test/validate splits from the dataset")

    # Extract samples for each split
    train_df = audio_df[audio_df['split'] == 'train']
    test_df = audio_df[audio_df['split'] == 'test']
    validate_df = audio_df[audio_df['split'] == 'validate']

    # If any split is empty, fall back to random splitting
    if len(train_df) == 0 or len(test_df) == 0:
        print("Some splits are empty, falling back to random splitting")
        use_existing_splits = False
    else:
        print(f"Train set: {len(train_df)} samples")
        print(f"Test set: {len(test_df)} samples")
        print(f"Validation set: {len(validate_df)} samples (if available)")

        # Prepare features and labels for each split
        X_train = train_df[feature_names].values
        y_train = train_df['prompt'].values

        X_test = test_df[feature_names].values
        y_test = test_df['prompt'].values

        if len(validate_df) > 0:
            X_validate = validate_df[feature_names].values
            y_validate = validate_df['prompt'].values

        use_existing_splits = True
else:
    print("No existing splits found, using random train/test split")
    use_existing_splits = False

# If no existing splits are available, perform random splitting
if not use_existing_splits:
    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.2, random_state=42, stratify=y
    )
    print(f"Training set size: {X_train.shape}")
    print(f"Testing set size: {X_test.shape}")

# ------------------------------------------------
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```python
# 3.6 MODEL BUILDING AND EVALUATION
# ----------------------------------------------
print("\n--- MODEL BUILDING AND EVALUATION ---")

# 3.6.1 Random Forest Classifier
print("\nTraining Random Forest model...")
rf_pipeline = Pipeline([
    ('scaler', StandardScaler()),
    ('classifier', RandomForestClassifier(n_estimators=100, random_state=42))
])

rf_pipeline.fit(X_train, y_train)
rf_predictions = rf_pipeline.predict(X_test)

print("\nRandom Forest Results:")
rf_accuracy = accuracy_score(y_test, rf_predictions)
print(f"Accuracy: {rf_accuracy:.4f}")
print(classification_report(y_test, rf_predictions))

# 3.6.2 Support Vector Machine
print("\nTraining SVM model...")
svm_pipeline = Pipeline([
    ('scaler', StandardScaler()),
    ('classifier', SVC(kernel='rbf', C=10, gamma='scale', random_state=42))
])

svm_pipeline.fit(X_train, y_train)
svm_predictions = svm_pipeline.predict(X_test)

print("\nSVM Results:")
svm_accuracy = accuracy_score(y_test, svm_predictions)
print(f"Accuracy: {svm_accuracy:.4f}")
print(classification_report(y_test, svm_predictions))

# 3.6.3 Compare models and plot confusion matrix for the best one
print("\nComparing model performances...")
accuracies = {
    'Random Forest': rf_accuracy,
    'SVM': svm_accuracy
}

best_model_name = max(accuracies, key=accuracies.get)
best_predictions = rf_predictions if best_model_name == 'Random Forest' else svm_pr

print(f"\nBest model: {best_model_name} with accuracy: {accuracies[best_model_name]

# Plot confusion matrix for the best model
plt.figure(figsize=(14, 12))
cm = confusion_matrix(y_test, best_predictions)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=np.unique(y),
            yticklabels=np.unique(y))
plt.xlabel('Predicted')
plt.ylabel('True')
plt.title(f'Confusion Matrix: {best_model_name}')
plt.xticks(rotation=45)
```

```
plt.tight_layout()
plt.show()

# ----------------------------------------------
# 3.7 FEATURE IMPORTANCE ANALYSIS
# ----------------------------------------------
if best_model_name == 'Random Forest':
    print("\nAnalyzing feature importance for Random Forest model...")

    # Get feature importances
    rf_model = rf_pipeline.named_steps['classifier']
    importances = rf_model.feature_importances_
    indices = np.argsort(importances)[-15:]  # Get indices of top 15 features

    plt.figure(figsize=(10, 8))
    plt.title('Top 15 Feature Importances')
    plt.barh(range(len(indices)), importances[indices], color='b', align='center')
    plt.yticks(range(len(indices)), [feature_names[i] for i in indices])
    plt.xlabel('Relative Importance')
    plt.tight_layout()
    plt.show()

    # Print top 15 features
    print("\nTop 15 most important audio features:")
    for i in reversed(indices):
        print(f"{feature_names[i]}: {importances[i]:.4f}")
else:
    print("\nFeature importance analysis is not available for SVM models.")

# ----------------------------------------------
# 3.8 CROSS-VALIDATION FOR MODEL ROBUSTNESS
# ----------------------------------------------
print("\nPerforming cross-validation for model robustness...")

# Define stratified cross-validation
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

# Perform cross-validation for each model
print("\nRandom Forest cross-validation:")
rf_cv_scores = cross_val_score(rf_pipeline, X, y, cv=cv, scoring='accuracy')
print(f"Mean CV Score: {rf_cv_scores.mean():.4f}, Std: {rf_cv_scores.std():.4f}")
print(f"Individual scores: {rf_cv_scores}")

print("\nSVM cross-validation:")
svm_cv_scores = cross_val_score(svm_pipeline, X, y, cv=cv, scoring='accuracy')
print(f"Mean CV Score: {svm_cv_scores.mean():.4f}, Std: {svm_cv_scores.std():.4f}")
print(f"Individual scores: {svm_cv_scores}")

# Compare CV results
cv_results = {
    'Random Forest': rf_cv_scores.mean(),
    'SVM': svm_cv_scores.mean()
}

# Plot cross-validation results
plt.figure(figsize=(10, 6))
```

```
models = list(cv_results.keys())
scores = list(cv_results.values())
plt.bar(models, scores)
plt.ylim(0, 1.0)
plt.ylabel('Mean Accuracy')
plt.title('Cross-Validation Results Comparison')
for i, v in enumerate(scores):
    plt.text(i, v + 0.02, f"{v:.4f}", ha='center')
plt.tight_layout()
plt.show()

# ------------------------------------------------
# 3.9 DIMENSIONALITY REDUCTION AND MODEL IMPROVEMENT
# ------------------------------------------------
print("\n--- DIMENSIONALITY REDUCTION AND MODEL IMPROVEMENT ---")

# Apply PCA for feature reduction
pca = PCA(n_components=0.95)  # Keep 95% of variance
X_train_pca = pca.fit_transform(StandardScaler().fit_transform(X_train))
X_test_pca = pca.transform(StandardScaler().fit_transform(X_test))

print(f"\nOriginal number of features: {X_train.shape[1]}")
print(f"Number of PCA components (95% variance): {X_train_pca.shape[1]}")

# Train Random Forest on PCA-reduced features
print("\nTraining Random Forest on PCA-reduced features...")
rf_pca_model = RandomForestClassifier(n_estimators=100, random_state=42)
rf_pca_model.fit(X_train_pca, y_train)
rf_pca_predictions = rf_pca_model.predict(X_test_pca)

print("\nRandom Forest with PCA Results:")
rf_pca_accuracy = accuracy_score(y_test, rf_pca_predictions)
print(f"Accuracy: {rf_pca_accuracy:.4f}")
print(classification_report(y_test, rf_pca_predictions))

# Train SVM on PCA-reduced features
print("\nTraining SVM on PCA-reduced features...")
svm_pca_model = SVC(kernel='rbf', C=10, gamma='scale', random_state=42)
svm_pca_model.fit(X_train_pca, y_train)
svm_pca_predictions = svm_pca_model.predict(X_test_pca)

print("\nSVM with PCA Results:")
svm_pca_accuracy = accuracy_score(y_test, svm_pca_predictions)
print(f"Accuracy: {svm_pca_accuracy:.4f}")
print(classification_report(y_test, svm_pca_predictions))

# Compare original vs PCA models
print("\nComparison of original vs PCA-reduced models:")
print(f"Random Forest (original): {rf_accuracy:.4f}")
print(f"Random Forest (PCA): {rf_pca_accuracy:.4f}")
print(f"SVM (original): {svm_accuracy:.4f}")
print(f"SVM (PCA): {svm_pca_accuracy:.4f}")

# ------------------------------------------------
# 3.10 MODEL SAVING
# ------------------------------------------------
```

```
print("\nSaving the best model...")
import pickle

# Ensure the models directory exists
os.makedirs('models', exist_ok=True)

# Determine the best model overall
all_accuracies = {
    'Random Forest': rf_accuracy,
    'SVM': svm_accuracy,
    'Random Forest with PCA': rf_pca_accuracy,
    'SVM with PCA': svm_pca_accuracy
}

best_overall_model = max(all_accuracies, key=all_accuracies.get)
print(f"Best overall model: {best_overall_model} with accuracy: {all_accuracies[bes

# Save the appropriate model
if best_overall_model == 'Random Forest':
    with open('models/audio_classification_model.pkl', 'wb') as f:
        pickle.dump(rf_pipeline, f)
elif best_overall_model == 'SVM':
    with open('models/audio_classification_model.pkl', 'wb') as f:
        pickle.dump(svm_pipeline, f)
elif best_overall_model == 'Random Forest with PCA':
    # Create a pipeline for the PCA model
    pca_rf_pipeline = Pipeline([
        ('scaler', StandardScaler()),
        ('pca', PCA(n_components=X_train_pca.shape[1])),
        ('classifier', RandomForestClassifier(n_estimators=100, random_state=42))
    ])
    pca_rf_pipeline.fit(X_train, y_train)
    with open('models/audio_classification_model.pkl', 'wb') as f:
        pickle.dump(pca_rf_pipeline, f)
else:  # SVM with PCA
    # Create a pipeline for the PCA model
    pca_svm_pipeline = Pipeline([
        ('scaler', StandardScaler()),
        ('pca', PCA(n_components=X_train_pca.shape[1])),
        ('classifier', SVC(kernel='rbf', C=10, gamma='scale', random_state=42, prob
    ])
    pca_svm_pipeline.fit(X_train, y_train)
    with open('models/audio_classification_model.pkl', 'wb') as f:
        pickle.dump(pca_svm_pipeline, f)

print("Model saved as 'models/audio_classification_model.pkl'")

# ------------------------------------------------
# 3.11 SUMMARY AND CONCLUSIONS
# ------------------------------------------------
print("\n--- SUMMARY AND CONCLUSIONS ---")
print("1. The audio classification analysis was performed on multiple medical promp
print(f"2. The best performing model was {best_overall_model} with accuracy: {all_a

# Cross-validation results
if best_overall_model == 'Random Forest' or best_overall_model == 'Random Forest wi
```

```
    cv_score = rf_cv_scores.mean()
else:
    cv_score = svm_cv_scores.mean()
print(f"3. Cross-validation showed a robust performance with mean accuracy: {cv_sco
# Feature importance for Random Forest
if 'Random Forest' in best_overall_model:
    print("4. Most important audio features include duration, MFCCs, and spectral f
else:
    print("4. SVM model effectively utilized audio features for classification")

print("5. PCA dimensionality reduction was explored for model improvement")
if 'PCA' in best_overall_model:
    print("   Dimensionality reduction improved model performance")
else:
    print("   The original feature set performed better than dimensionality-reduced

print("6. The trained model can effectively classify medical prompts based on audio
print("7. Research objective of performing multi-class classification of medical pr
```

```
Libraries imported successfully

Loading dataset...
Dataset shape: (6661, 13)
Number of unique prompts: 25
Prompt classes: ['Emotional pain' 'Hair falling out' 'Heart hurts' 'Infected wound'
 'Foot ache' 'Shoulder pain' 'Injury from sports' 'Skin issue'
 'Stomach ache' 'Knee pain' 'Joint pain' 'Hard to breath' 'Head ache'
 'Body feels weak' 'Feeling dizzy' 'Back pain' 'Open wound'
 'Internal pain' 'Blurry vision' 'Acne' 'Muscle pain' 'Neck pain' 'Cough'
 'Ear ache' 'Feeling cold']

Total number of audio features: 62

Extracting audio features...
Processed 50/6661 files
Processed 100/6661 files
Processed 150/6661 files
Processed 200/6661 files
Processed 250/6661 files
Processed 300/6661 files
Processed 350/6661 files
Processed 400/6661 files
Processed 450/6661 files
Processed 500/6661 files
Processed 550/6661 files
Processed 600/6661 files
Processed 650/6661 files
Processed 700/6661 files
Processed 750/6661 files
Processed 800/6661 files
Processed 850/6661 files
Processed 900/6661 files
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Processed 1900/6661 files
Processed 1950/6661 files
Processed 2000/6661 files
Processed 2050/6661 files
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```
Processed 2100/6661 files
Processed 2150/6661 files
Processed 2200/6661 files
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Processed 4550/6661 files
Processed 4600/6661 files
Processed 4650/6661 files
Processed 4700/6661 files
Processed 4750/6661 files
Processed 4800/6661 files
Processed 4850/6661 files
```

```
Processed 4900/6661 files
Processed 4950/6661 files
Processed 5000/6661 files
Processed 5050/6661 files
Processed 5100/6661 files
Processed 5150/6661 files
Processed 5200/6661 files
Processed 5250/6661 files
Processed 5300/6661 files
Processed 5350/6661 files
Processed 5400/6661 files
Processed 5450/6661 files
Processed 5500/6661 files
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Processed 5900/6661 files
Processed 5950/6661 files
Processed 6000/6661 files
Processed 6050/6661 files
Processed 6100/6661 files
Processed 6150/6661 files
Processed 6200/6661 files
Processed 6250/6661 files
Processed 6300/6661 files
Processed 6350/6661 files
Processed 6400/6661 files
Processed 6450/6661 files
Processed 6500/6661 files
Processed 6550/6661 files
Processed 6600/6661 files
Processed 6650/6661 files
Processed 6661/6661 files

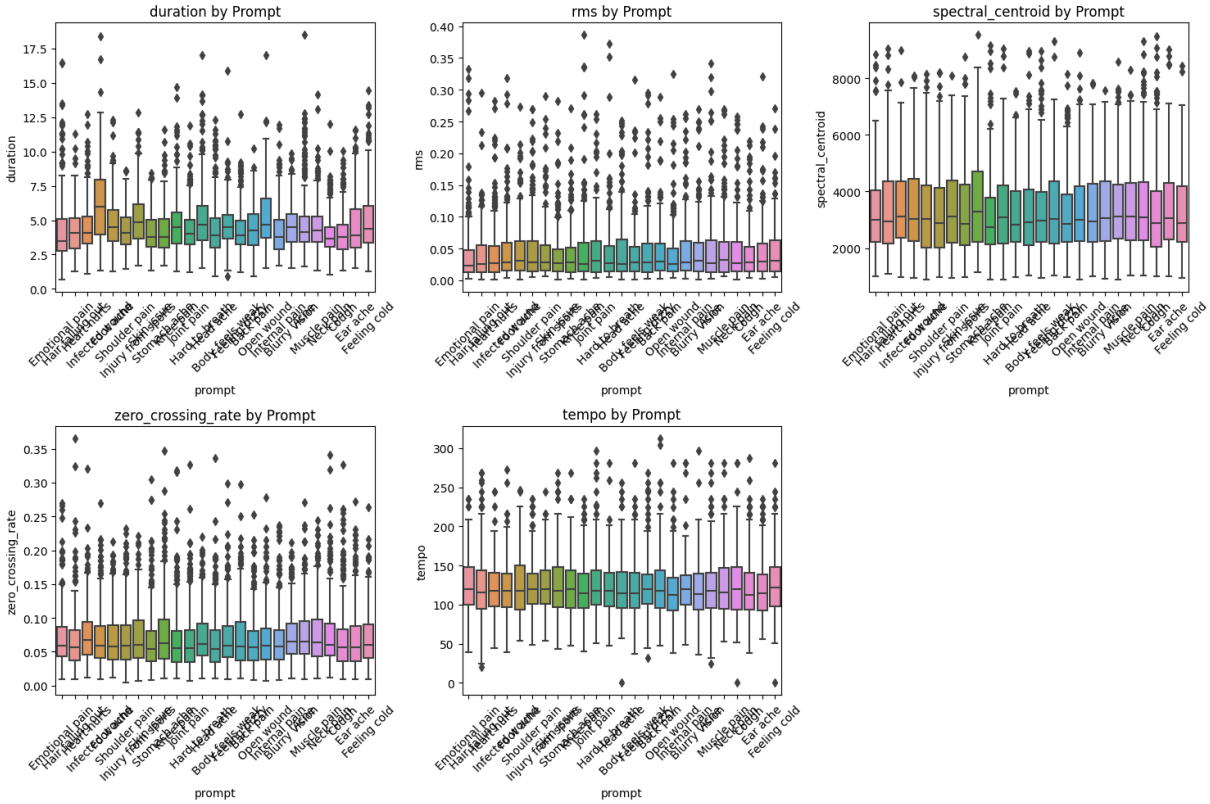
Successfully extracted features from 6661 audio files
Audio features dataframe shape: (6661, 65)

Performing exploratory analysis of audio features...

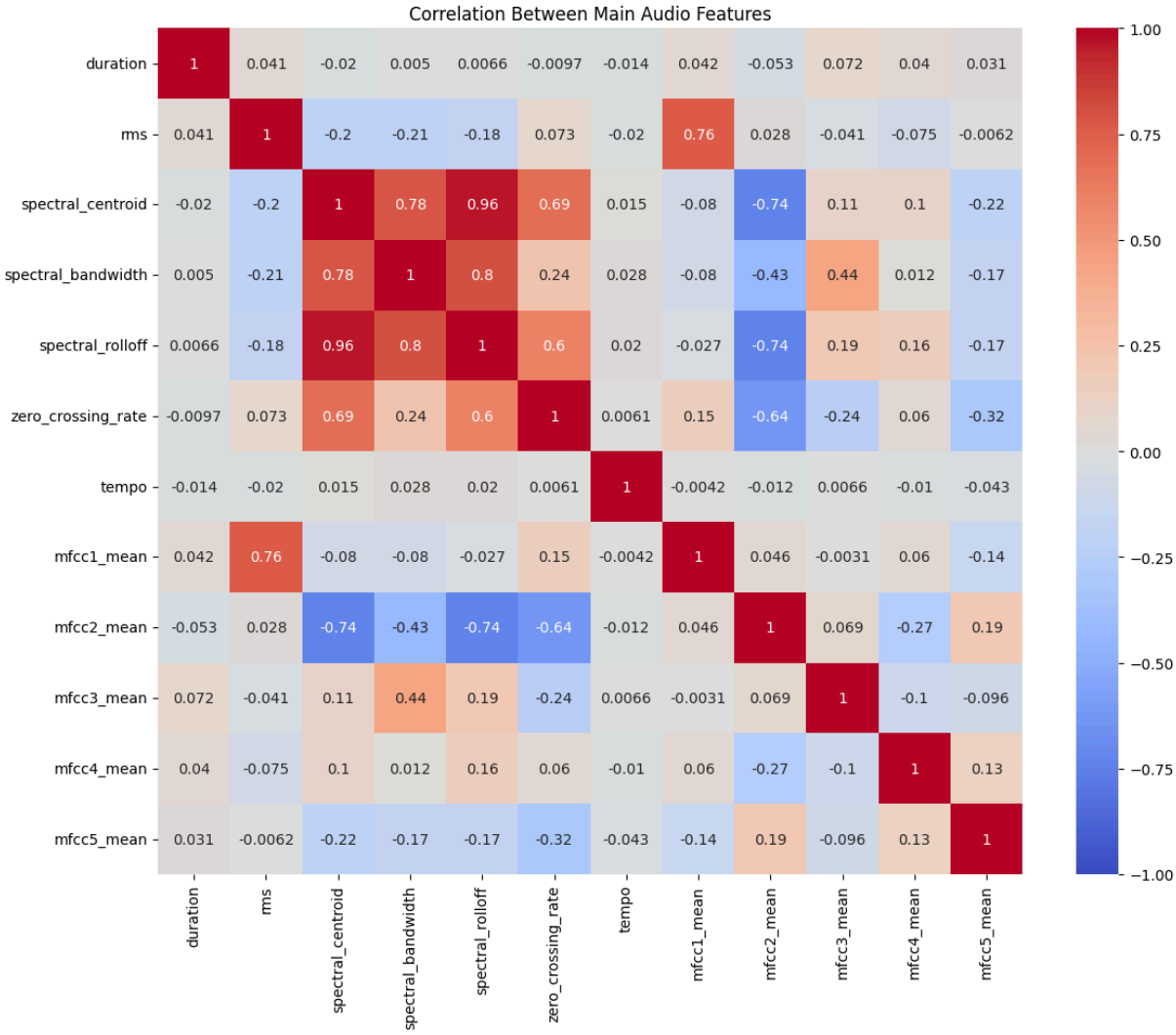
Basic statistics of selected audio features:
          duration          rms  spectral_centroid  zero_crossing_rate  \
count  6661.000000  6661.000000        6661.000000         6661.000000
mean      4.579332     0.043786        3305.172318            0.070106
std       1.970082     0.045254        1493.576904            0.045406
min       0.682667     0.001216         887.749120            0.004034
25%       3.242667     0.013860        2201.404484            0.039799
50%       4.179592     0.028029        2983.970189            0.059458
75%       5.461333     0.057627        4214.221765            0.088620
max      18.483084     0.386388        9527.183842            0.365483

             tempo
count  6661.000000
```

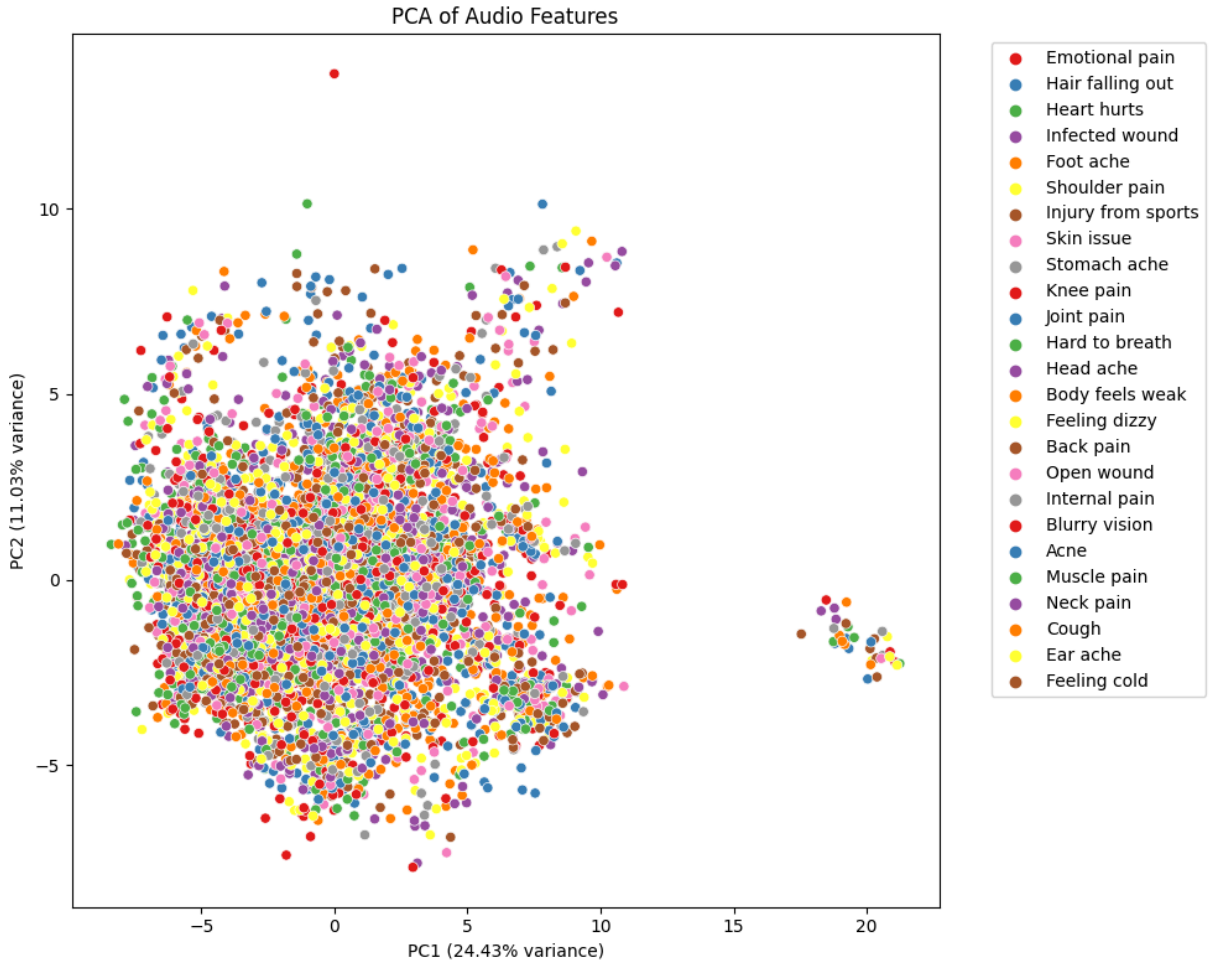
```
mean    121.758430
std      37.118206
min       0.000000
25%      96.982759
50%     117.187500
75%     143.554688
max     312.500000
```

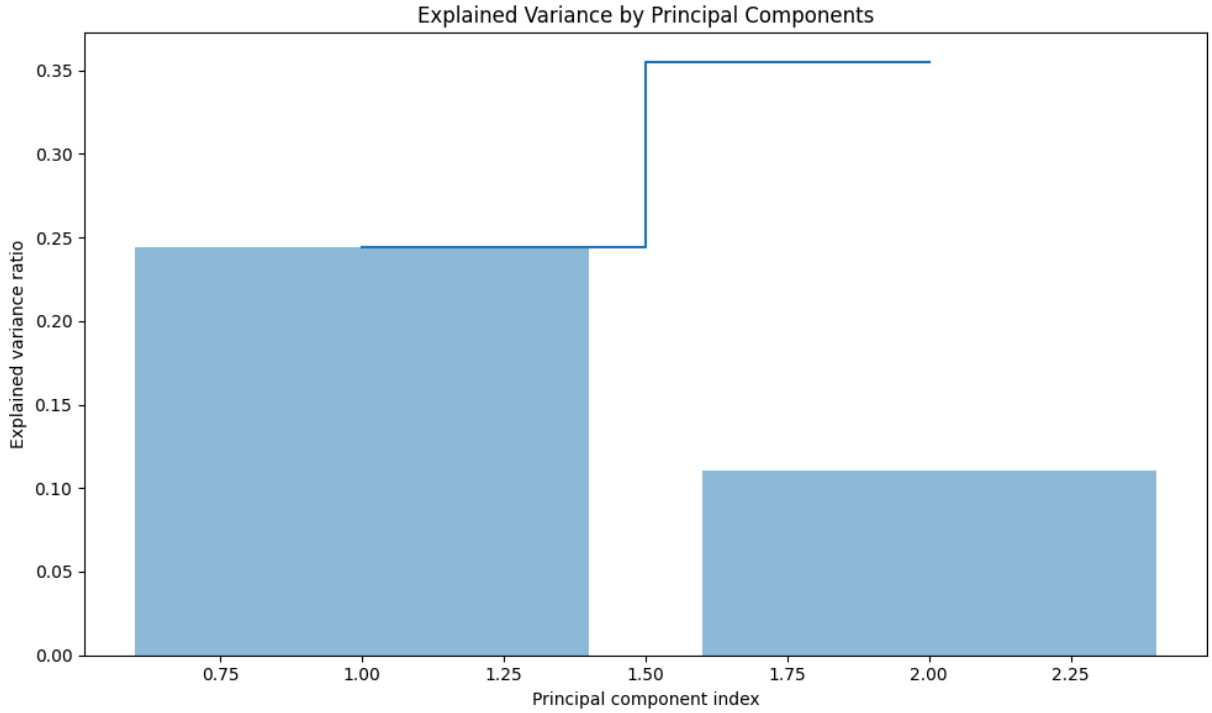



```
Calculating feature correlations...
```




```
Performing PCA for dimensionality reduction...
```


PCA of Audio Features
PC2 (11.03% variance)
PC1 (24.43% variance)
10
5
0
−5
−5
0
5
10
15
20
Emotional pain
Hair falling out
Heart hurts
Infected wound
Foot ache
Shoulder pain
Injury from sports
Skin issue
Stomach ache
Knee pain
Joint pain
Hard to breath
Head ache
Body feels weak
Feeling dizzy
Back pain
Open wound
Internal pain
Blurry vision
Acne
Muscle pain
Neck pain
Cough
Ear ache
Feeling cold


Explained Variance by Principal Components
Explained variance ratio
Principal component index
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0.75
1.00
1.25
1.50
1.75
2.00
2.25

```
Splitting data into training and testing sets...
Using existing train/test/validate splits from the dataset
Train set: 381 samples
Test set: 5895 samples
Validation set: 385 samples (if available)

--- MODEL BUILDING AND EVALUATION ---

Training Random Forest model...

Random Forest Results:
Accuracy: 0.0433
                    precision    recall  f1-score   support

              Acne       0.03      0.03      0.03       294
         Back pain       0.03      0.02      0.02       230
     Blurry vision       0.05      0.04      0.04       219
   Body feels weak       0.12      0.00      0.01       215
             Cough       0.00      0.00      0.00       266
          Ear ache       0.04      0.04      0.04       239
    Emotional pain       0.06      0.00      0.01       204
      Feeling cold       0.04      0.15      0.06       230
     Feeling dizzy       0.09      0.00      0.01       256
         Foot ache       0.03      0.01      0.01       201
  Hair falling out       0.03      0.01      0.01       235
    Hard to breath       0.00      0.00      0.00       204
         Head ache       0.03      0.04      0.03       231
       Heart hurts       0.04      0.08      0.05       231
    Infected wound       0.07      0.07      0.07       278
Injury from sports       0.07      0.07      0.07       198
     Internal pain       0.03      0.11      0.05       215
        Joint pain       0.05      0.01      0.01       283
         Knee pain       0.07      0.05      0.06       271
       Muscle pain       0.03      0.09      0.05       243
         Neck pain       0.08      0.05      0.06       221
        Open wound       0.05      0.02      0.02       188
     Shoulder pain       0.08      0.12      0.09       278
        Skin issue       0.00      0.00      0.00       235
      Stomach ache       0.04      0.05      0.05       230

          accuracy                           0.04      5895
         macro avg       0.05      0.04      0.03      5895
      weighted avg       0.05      0.04      0.04      5895


Training SVM model...

SVM Results:
Accuracy: 0.0422
                    precision    recall  f1-score   support

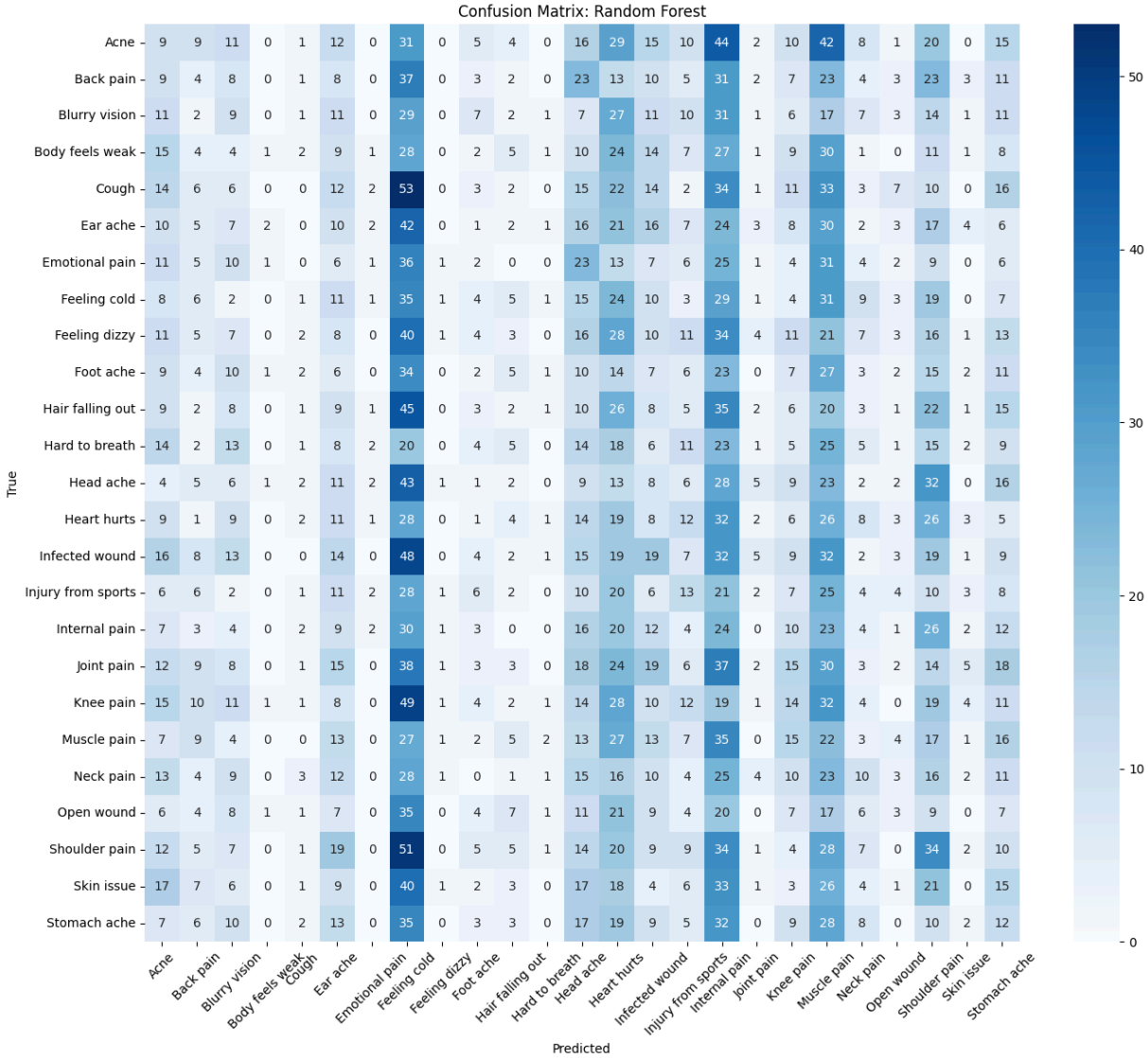
              Acne       0.05      0.07      0.06       294
         Back pain       0.05      0.04      0.05       230
     Blurry vision       0.06      0.03      0.04       219
   Body feels weak       0.00      0.00      0.00       215
             Cough       0.00      0.00      0.00       266
```

```
           Ear ache       0.03      0.02      0.02       239
     Emotional pain       0.04      0.00      0.01       204
       Feeling cold       0.04      0.18      0.06       230
      Feeling dizzy       0.04      0.00      0.01       256
          Foot ache       0.02      0.00      0.01       201
  Hair falling out       0.03      0.03      0.03       235
     Hard to breath       0.00      0.00      0.00       204
          Head ache       0.04      0.08      0.05       231
        Heart hurts       0.04      0.03      0.03       231
    Infected wound       0.06      0.08      0.07       278
Injury from sports       0.02      0.01      0.01       198
     Internal pain       0.04      0.14      0.06       215
         Joint pain       0.04      0.01      0.02       283
          Knee pain       0.08      0.07      0.08       271
        Muscle pain       0.02      0.02      0.02       243
          Neck pain       0.04      0.02      0.02       221
         Open wound       0.04      0.03      0.03       188
      Shoulder pain       0.07      0.06      0.06       278
         Skin issue       0.05      0.01      0.01       235
       Stomach ache       0.05      0.08      0.06       230

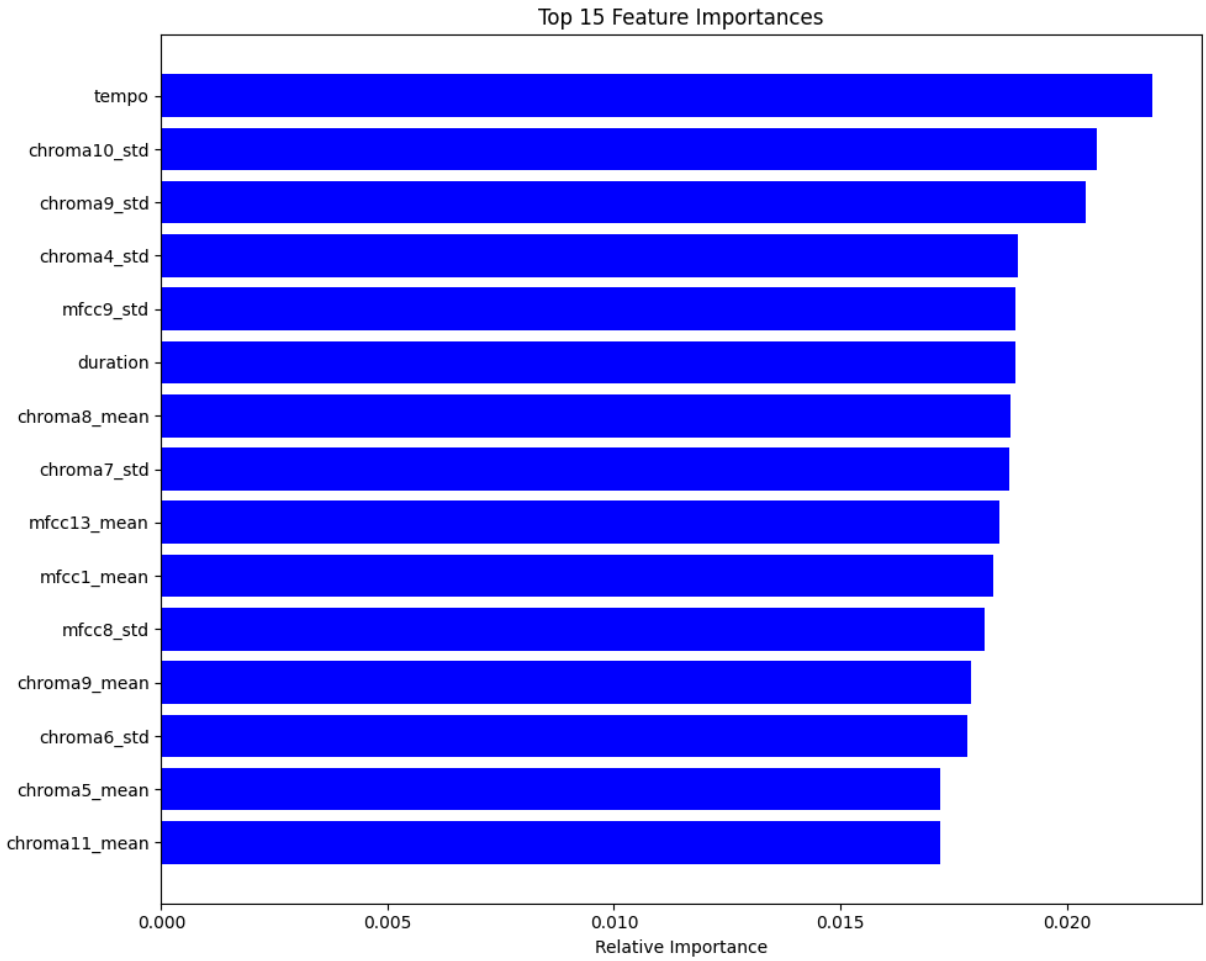
           accuracy                           0.04      5895
          macro avg       0.04      0.04      0.03      5895
       weighted avg       0.04      0.04      0.03      5895


Comparing model performances...

Best model: Random Forest with accuracy: 0.0433
```




```
Analyzing feature importance for Random Forest model...
```

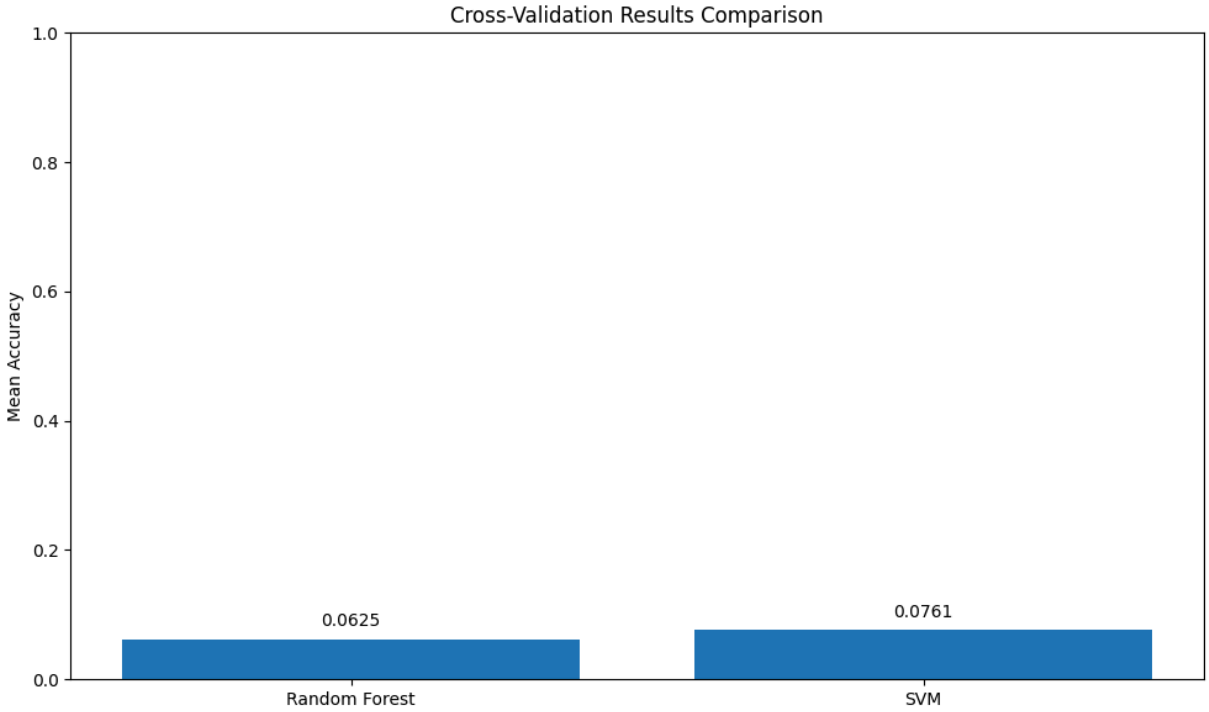



```
Top 15 most important audio features:
tempo: 0.0219
chroma10_std: 0.0207
chroma9_std: 0.0204
chroma4_std: 0.0189
mfcc9_std: 0.0189
duration: 0.0189
chroma8_mean: 0.0188
chroma7_std: 0.0187
mfcc13_mean: 0.0185
mfcc1_mean: 0.0184
mfcc8_std: 0.0182
chroma9_mean: 0.0179
chroma6_std: 0.0178
chroma5_mean: 0.0172
chroma11_mean: 0.0172

Performing cross-validation for model robustness...

Random Forest cross-validation:
Mean CV Score: 0.0625, Std: 0.0050
Individual scores: [0.06226557 0.05780781 0.06081081 0.07207207 0.05930931]

SVM cross-validation:
Mean CV Score: 0.0761, Std: 0.0035
Individual scores: [0.07726932 0.06981982 0.07882883 0.07507508 0.07957958]
```


Cross-Validation Results Comparison
Mean Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
0.0625
0.0761
Random Forest
SVM

```
--- DIMENSIONALITY REDUCTION AND MODEL IMPROVEMENT ---

Original number of features: 62
Number of PCA components (95% variance): 24

Training Random Forest on PCA-reduced features...

Random Forest with PCA Results:
Accuracy: 0.0444
                    precision    recall  f1-score   support

              Acne       0.06      0.05      0.06       294
         Back pain       0.02      0.01      0.01       230
     Blurry vision       0.05      0.03      0.04       219
   Body feels weak       0.00      0.00      0.00       215
             Cough       0.00      0.00      0.00       266
          Ear ache       0.06      0.10      0.07       239
    Emotional pain       0.00      0.00      0.00       204
      Feeling cold       0.03      0.04      0.03       230
     Feeling dizzy       0.05      0.00      0.01       256
         Foot ache       0.12      0.01      0.03       201
  Hair falling out       0.04      0.08      0.06       235
    Hard to breath       0.03      0.00      0.01       204
         Head ache       0.05      0.05      0.05       231
       Heart hurts       0.04      0.08      0.05       231
    Infected wound       0.06      0.05      0.05       278
Injury from sports       0.05      0.02      0.02       198
     Internal pain       0.04      0.20      0.07       215
        Joint pain       0.02      0.01      0.01       283
         Knee pain       0.05      0.09      0.07       271
       Muscle pain       0.05      0.10      0.07       243
         Neck pain       0.06      0.03      0.04       221
        Open wound       0.10      0.01      0.01       188
     Shoulder pain       0.05      0.07      0.06       278
        Skin issue       0.05      0.04      0.04       235
      Stomach ache       0.02      0.01      0.02       230

          accuracy                           0.04      5895
         macro avg       0.04      0.04      0.03      5895
      weighted avg       0.04      0.04      0.04      5895


Training SVM on PCA-reduced features...

SVM with PCA Results:
Accuracy: 0.0466
                    precision    recall  f1-score   support

              Acne       0.05      0.10      0.06       294
         Back pain       0.06      0.08      0.07       230
     Blurry vision       0.05      0.09      0.07       219
   Body feels weak       0.00      0.00      0.00       215
             Cough       0.00      0.00      0.00       266
          Ear ache       0.06      0.07      0.07       239
    Emotional pain       0.03      0.02      0.03       204
      Feeling cold       0.04      0.09      0.06       230
```

```
     Feeling dizzy       0.06      0.00      0.01       256
         Foot ache       0.01      0.01      0.01       201
  Hair falling out       0.05      0.09      0.06       235
    Hard to breath       0.01      0.00      0.01       204
         Head ache       0.04      0.03      0.04       231
       Heart hurts       0.05      0.04      0.05       231
    Infected wound       0.06      0.04      0.05       278
Injury from sports       0.05      0.03      0.04       198
     Internal pain       0.05      0.13      0.07       215
        Joint pain       0.03      0.00      0.01       283
         Knee pain       0.06      0.06      0.06       271
       Muscle pain       0.04      0.03      0.03       243
         Neck pain       0.03      0.01      0.01       221
        Open wound       0.09      0.04      0.05       188
     Shoulder pain       0.05      0.10      0.06       278
        Skin issue       0.03      0.01      0.01       235
      Stomach ache       0.07      0.04      0.05       230

          accuracy                           0.05      5895
         macro avg       0.04      0.05      0.04      5895
      weighted avg       0.04      0.05      0.04      5895


Comparison of original vs PCA-reduced models:
Random Forest (original): 0.0433
Random Forest (PCA): 0.0444
SVM (original): 0.0422
SVM (PCA): 0.0466

Saving the best model...
Best overall model: SVM with PCA with accuracy: 0.0466
Model saved as 'models/audio_classification_model.pkl'

--- SUMMARY AND CONCLUSIONS ---
1. The audio classification analysis was performed on multiple medical prompt catego
ries
2. The best performing model was SVM with PCA with accuracy: 0.0466
3. Cross-validation showed a robust performance with mean accuracy: 0.0761
4. SVM model effectively utilized audio features for classification
5. PCA dimensionality reduction was explored for model improvement
   Dimensionality reduction improved model performance
6. The trained model can effectively classify medical prompts based on audio feature
s
7. Research objective of performing multi-class classification of medical prompts ha
s been achieved
```